

Computer Vision Quiz

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1. Why can a ResNet be trained successfully with 152+ layers, while stacking plain convolutional layers beyond ~20 degrades performance? 1 point



A. Deeper plain networks have too few parameters to fit the data



B. Skip connections act as identity shortcuts that guarantee gradient flow through the network



C. ResNet uses larger convolution kernels that capture more context



D. ResNet removes all pooling layers, preserving spatial resolution

Clear selection

2. In transfer learning with a pretrained ResNet-50, what is the purpose of freezing the backbone and replacing only the final fully-connected head? 1 point



A. It reduces GPU memory so larger images can be used



B. It forces the network to relearn low-level features for the new dataset



C. It preserves the learned ImageNet features while adapting the output to the new classes



D. It disables Batch Normalisation, which is unnecessary for small datasets

Clear selection



3. How does Batch Normalisation behave differently at inference time compared to training time?

1 point

- ☒ A. It switches from batch statistics to running mean and variance accumulated during training
- ☐ B. It normalises using the statistics of the current test batch
- ☐ C. It is removed entirely from the computation graph
- ☐ D. It recomputes statistics from the full training set on every forward pass

Clear selection

4. When implementing GradCAM for a ResNet-50, on which layer should the forward hook be registered to capture activations?

1 point

- ☐ A. The first convolutional layer
- ☐ B. The average-pooling layer
- ☐ C. The final fully-connected (fc) layer
- ☒ D. The final convolutional layer (e.g., `model.layer4[-1]`)

Clear selection

5. What key component does Faster R-CNN introduce to replace the slow selective-search proposal step used in earlier R-CNN variants?

1 point

- ☒ A. A Region Proposal Network (RPN) that predicts objectness scores and box deltas from anchor boxes
- ☐ B. Non-Maximum Suppression applied before the backbone
- ☐ C. A Transformer decoder that outputs a fixed set of boxes
- ☐ D. Focal loss applied to every pixel of the feature map

Clear selection

6. RetinaNet introduces focal loss, $FL(p_t) = -(1 - p_t)^\gamma \cdot \log(p_t)$. What problem does it solve for one-stage detectors?

1 point

- ☐ A. Vanishing gradients in very deep backbones
- ☐ B. Overlapping boxes producing duplicate detections
- ☒ C. The extreme foreground-background class imbalance, by down-weighting easy negative examples
- ☐ D. Poor localisation of small objects due to coarse feature maps

Clear selection

7. For box A = [10, 20, 50, 80] and box B = [30, 40, 70, 100] in (x1, y1, x2, y2) format, what is the IoU?

1 point

- ☐ A. 0.10
- ☒ B. 0.20
- ☐ C. 0.33
- ☐ D. 0.50

Clear selection

8. Why does Faster R-CNN use RoI Align instead of RoI Pooling?

1 point

- ☐ A. RoI Align runs faster on GPU hardware
- ☒ B. RoI Align avoids the quantisation artefact of snapping region boundaries to the feature-map grid, preserving spatial precision
- ☐ C. RoI Align eliminates the need for anchor boxes
- ☐ D. RoI Align outputs variable-sized features for each proposal

Clear selection

9. Two boxes of the same product sit side by side on a conveyor belt. Which model can label them as two separate object instances? 1 point

- ☐ A. FCN, because it replaces fully-connected layers with 1×1 convolutions
- ☐ B. DeepLabv3+, because it uses atrous spatial pyramid pooling
- ☒ C. Mask R-CNN, because instance segmentation produces a separate mask per detected object
- ☐ D. Any semantic segmentation model, since per-pixel labels imply per-object masks

Clear selection

10. What is the role of skip connections in the U-Net architecture? 1 point

- ☐ A. They allow the network to skip training on easy examples
- ☒ B. They concatenate encoder feature maps to the decoder at each resolution, preserving fine spatial detail lost in downsampling
- ☐ C. They add residual identities that stabilise very deep encoders
- ☐ D. They pass the input image directly to the output for faster convergence

Clear selection

11. Using $k_{\text{eff}} = k + (k - 1)(r - 1)$, what is the effective kernel size of a 3×3 atrous convolution with dilation rate $r = 6$? 1 point

- ☐ A. 9
- ☐ B. 11
- ☒ C. 13
- ☐ D. 18

Clear selection

12. Which types of prompts can the Segment Anything Model (SAM) accept to produce a mask? 1 point

- ☐ A. Only a full ground-truth mask of the target object
- ☒ B. Point, box, or text prompts
- ☐ C. Only class labels from the COCO vocabulary
- ☐ D. A morphological structuring element and kernel size

Clear selection

13. In a VAE, why is the reparameterisation trick $z = \mu + \sigma \cdot \epsilon$ (with $\epsilon \sim N(0,1)$) required? 1 point

- ☐ A. It reduces the number of latent dimensions needed
- ☒ B. Sampling z directly from $N(\mu, \sigma^2)$ is not differentiable; moving the randomness into ϵ lets gradients flow through μ and σ
- ☐ C. It guarantees the decoder output stays within the range $[0, 1]$
- ☐ D. It prevents the discriminator from overpowering the generator

Clear selection

14. If the KL term of the ELBO loss is disabled by setting $\beta = 0$, what happens to the VAE? 1 point

- ☐ A. Reconstructions become blurrier because the decoder capacity is reduced
- ☐ B. Training diverges because the reconstruction term alone is unstable
- ☒ C. The latent space loses its smooth structure aligned with the $N(0,1)$ prior – the model degenerates toward a plain autoencoder
- ☐ D. The model collapses to producing a single output for all inputs

Clear selection

15. Why are VAE reconstructions characteristically blurry while GAN outputs are sharp? 1 point

- ☒ A. The VAE's MSE reconstruction term averages over all valid decodings, producing a blurred mean image
- ☐ B. VAEs use smaller networks than GANs by design
- ☐ C. The KL divergence term removes high-frequency detail from images
- ☐ D. VAEs are trained on downsampled images, so detail is unrecoverable

Clear selection

16. What does the Fréchet Inception Distance (FID) measure when evaluating generative models? 1 point

- ☐ A. The pixel-wise MSE between each generated image and its nearest real image
- ☐ B. The classification accuracy of InceptionV3 on generated images
- ☐ C. The percentage of generated images a human judge labels as real
- ☒ D. The distance between the InceptionV3 feature distributions of real and generated images – lower is better

Clear selection

17. For a ViT-B/16 processing a 224×224 image, what is the total sequence length entering the Transformer encoder, including the [CLS] token? 1 point

- ☐ A. 196
- ☒ B. 197
- ☐ C. 256
- ☐ D. 785

Clear selection

18. How does CLIP classify CIFAR-10 images zero-shot, without ever training on CIFAR-10? 1 point

- ☐ A. It fine-tunes its final layer on 500 CIFAR-10 images at inference time
- ☐ B. It uses a k-nearest-neighbour lookup over its 400M training images
- ☒ C. It compares each image embedding to text-prompt embeddings (e.g., 'a photo of a dog') in a shared image-text embedding space and picks the most similar
- ☐ D. It generates a caption for the image and string-matches it to class names

Clear selection

19. What is the main purpose of shifted windows in the Swin Transformer? 1 point

- ☒ A. To enable local windowed attention that avoids the quadratic cost of global attention while still allowing cross-window information flow
- ☐ B. To randomly augment training patches for better generalisation
- ☐ C. To replace positional embeddings entirely
- ☐ D. To convert the ViT into a purely convolutional network

Clear selection

20. QuickVision's Jetson Orin cameras have a 33ms per-frame budget at 30fps. Given ResNet-50 $\approx$ 15ms, ViT-B/16 $\approx$ 30ms, and Faster R-CNN $\approx$ 200ms, which model cannot run in real time? 1 point

- ☐ A. ResNet-50
- ☐ B. ViT-B/16
- ☒ C. Faster R-CNN
- ☐ D. All three fit within the budget

Clear selection

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